Applied Statistics

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Outline

1. Multivariate normal data.
2. Inference for Gaussian models.
3. Principal component analysis.
4. Factor analysis.
5. Classification.
6. Cluster methods.
7. Resampling methods.
8. The general linear model.
9. Regression models.
10. Analysis of variance.
11. Multifactor designs with no interactions.
12. Multifactor designs with repeated measurements.

Chapter 1

Chapter 1

1. Random vectors.

    1.1. Multivariate Data.

    1.3. Expectation and variance.

    1.4. Estimation of μ and Σ.

    1.4. The multivariate normal distribution.

    1.5. Fisher's lemma.

Multivariate Data

In this course we will be very often concerned with multivariate data, i.e. data with several outcome variables.

Example: in a medical study p different blood values of patients suffering a certain disease are analysed.

The first two chapters of this course develop some background on the multivariate extensions of basic univariate statistical tools and methods for such type of data.

Multivariate Data

When we talk about an observation, we mean two different things:

(1) The actual numbers measured:

$$x = (x_1, \ldots, x_p)' = (0.3, 6.4, 1.1)'$$

(2) The model behind the data:

$$X = (X_1, \ldots, X_p)'$$

$\Longrightarrow$ a random vector. (Throughout we use the convention that a data point is written as a column vector.)

Multivariate Data

Consider a generic p-vector X. A general target in statistics is to infer the distribution of X from a sample of measurements.

Joint distribution:

$$P(X \in (a, b]) = P(X_i \in (a_i, b_i], \quad 1 \leq i \leq p).$$

Would we know the joint density $f(x) = f(x_1, \ldots, x_p)$ of X:

$$P(X \in (a, b]) = \int_{a_1}^{b_1} \ldots \int_{a_p}^{b_p} f(x_1, \ldots, x_p) dx_1 \ldots dx_p.$$

Note: Generally the joint distribution is not determined by the marginals!

Multivariate Data

How could we get the density/distribution from data (sample of size n)?

- Parametric estimation: $f(x) = f(x|\theta) \Longrightarrow$ estimate θ, analog the same line as in the univariate case. There are not that many parametric densities for multivariate data ...
- Non-parametric estimation: for small h

$$
\begin{aligned}
f(x) &\approx \frac{1}{\mathrm{Vol}((x, x+h])} \times \int_{(x,x+h]} f(t)dt \\
&\approx \frac{1}{\mathrm{Vol}((x, x+h])} \times \frac{\text{number of samples in } (x, x+h]}{n}.
\end{aligned}
$$

If $p > 1$, then there are not many samples in $(x, x+h]$ ...

Expectation and variance

Very often, it is too difficult to infer the distribution. Then often we focus on mean and variance.

Then

$$\mu = EX := (EX_1, \ldots, EX_p)'$$

is called the expectation of X. (Componentwise expectations.)

The $p \times p$ matrix

$$\Sigma := \mathrm{Var}(X) = E(X - \mu)(X - \mu)'$$

is called (variance)-covariance matrix of X.

Expectation and variance

Recall: The expected value of a scalar variable X with density function $f(x)$ is

$$EX = \int_{-\infty}^{\infty} xf(x)dx.$$

Denoting $f(x, y)$ the joint density of X and Y, then the covariance is

$$\begin{aligned}\mathrm{Cov}(X, Y) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x - EX)(y - EY)f(x, y)dxdy \\ &= \int_{-\infty}^{\infty} xyf(x, y)dxdy - EXEY.\end{aligned}$$

Expectation and variance

The matrix Σ plays a very important role in several statistical procedures. We will need also theoretical properties of it. The most important are:

- Σ is symmetric: $\Sigma = \Sigma'$.
- Σ is non-negative definite: for all $v \in \mathbb{R}^p$ we have $v'\Sigma v \geq 0$.

$$\begin{aligned} v'\Sigma v &= v'E(X-\mu)(X-\mu)'v. \\ &= Ev'(X-\mu)(X-\mu)'v. \\ &= E|v'(X-\mu)|^2 \geq 0. \end{aligned}$$

- $\Sigma = EXX' - \mu\mu'$.

Expectation and variance

Using $|\Sigma_{ij}| \leq |\Sigma_{ii}|^{1/2}|\Sigma_{jj}|^{1/2}$ we get

$$\rho_{ij} := \frac{\Sigma_{ij}}{\Sigma_{ii}^{1/2}\Sigma_{jj}^{1/2}} \in [-1, 1].$$

The matrix $K = ((\rho_{ij}))$ is called correlation matrix of X.

Henceforth we tacitly impose that second order moments exist.

Expectation and variance

Let X be a p-r.v. and Y be q-r.v. Then

$$\mathrm{Cov}(X, Y) := E(X - \mu_X)(Y - \mu_Y)'$$

is called the covariance (matrix) between X and Y.

Some useful rules:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + \underbrace{\mathrm{Cov}(X, Y) + \mathrm{Cov}(Y, X)}_{\text{generally not identical!}}$$

$$\mathrm{Cov}(X, Y) = EXY' - \mu_X \mu_Y'.$$

Expectation and variance

Very often we will apply affine linear mappings to our data:
$X \mapsto AX + b$ (e.g. rescaling, shifting, whitening, rotating, etc.)

Some useful rules:

$$\mathrm{Var}(AX + b) = A\Sigma A';$$
$$\mathrm{Cov}(AX + a, BY + b) = A\mathrm{Cov}(X, Y)B';$$
$$\mathrm{Cov}(A_1X_1 + A_2X_2, Y) = A_1\mathrm{Cov}(X_1, Y) + A_2\mathrm{Cov}(X_2, Y).$$

Multivariate Data

Let's switch now from population to sample.

We have a sample $X_1, \ldots, X_n$ where $X_i = (X_{i1}, \ldots, X_{ip})'$. (Please mind the abuse of notation! It should be clear from the context.)

The data matrix is an $n \times p$ matrix

$$X = (X_1, \ldots, X_n)'.$$

Multivariate Data

When we look at a realisation of our rv's we use lower case characters.

	variable 1	variable 2	variable p
x_1'	2.33	-5.30	2.20
x_2'	7.49	-2.46	1.92
x_3'	7.09	2.40	1.54
$\ldots$	$\ldots$	$\ldots$	$\ldots$
$\ldots$	$\ldots$	$\ldots$	$\ldots$
x_n'	1.58	0.56	3.06

Estimation of μ and Σ

Let $X_1, \ldots, X_n$ be i.i.d. p-r.v.s with finite first moment.

Then a common estimator for $\mu = EX_1$ is the sample (empirical) mean

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Proposition

(i) This estimator is unbiased: $E[\bar{X}] = \mu$.

(ii) $\mathrm{Var}[\bar{X}] = \frac{1}{n}\Sigma$. It follows that $\bar{X} \xrightarrow{P} \mu$. Hence the estimator is consistent.

Estimation of μ and Σ

Let $X_1, \ldots, X_n$ be i.i.d. p-r.v.s with 2nd moments. A common estimator of $\Sigma = \mathrm{Var}(X_1) = E(X_1 - EX_1)(X_1 - EX_1)'$ is the sample (empirical) covariance matrix

$$S = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(X_i - \bar{X})'.$$

Proposition

$E[S] = \Sigma$ and $S \xrightarrow{P} \Sigma$. Hence the estimator is unbiased and consistent.

Proof. Exercise.

The sample (empirical) correlation matrix is denoted by $R = ((R_{ij}))$ with $R_{ij} = \frac{S_{ij}}{\sqrt{S_{ii}}\sqrt{S_{jj}}}$.

Estimation of μ and Σ

Here a little numerical illustration. We have sample $x_1, \ldots, x_{100}$ with $x_i \in \mathbb{R}^3$. The data are relizations of a distribution with

$$\mu = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 17.8 & -3.0 & -1.9 \\ -3.1 & 15.2 & 5.6 \\ -1.9 & 5.6 & 5.7 \end{pmatrix}$$

	variable 1	variable 2	variable 3
...	...	...	...
...	...	...	...
x_{98}	0.94	-0.40	2.59
x_{99}	6.69	9.03	-0.35
x_{100}	3.33	-2.43	-2.33
$\bar{\mathbf{x}}$	**4.71**	**3.44**	**1.08**

Estimation of μ and Σ

For the covariance we obtain

$$S = \begin{pmatrix} 14.5 & -2.6 & -1.3 \\ -2.6 & 13.3 & 3.9 \\ -1.3 & 3.9 & 5.0 \end{pmatrix} \quad \left(\Sigma = \begin{pmatrix} 17.8 & -3.0 & -1.9 \\ -3.0 & 15.2 & 5.6 \\ -1.9 & 5.6 & 5.7 \end{pmatrix} \right)$$

and

$$R = \begin{pmatrix} 1.0 & -0.2 & -0.1 \\ -0.2 & 1.00 & 0.5 \\ -0.1 & 0.5 & 1.00 \end{pmatrix}.$$

Estimation of μ and Σ

Note that the rules for population mean and covariance matrices extend to their empirical versions. E.g. if $Y_i = AX_i + b$ then

$$\overline{Y} = A\overline{X} + b,$$

and

$$S^Y = AS^X A'.$$

Multivariate normal distribution

For multivariate data, we don't have a big variety of models available.

A simple idea is to start with a vector $Z = (Z_1, \ldots, Z_q)'$ having independent components and then to apply some mapping

$$Z \mapsto X = A(Z) = (a_1(Z), \ldots, a_p(Z))'$$

in order to obtain a random vector, where the components now are dependent.

This is exactly the idea behind the multivariate normal distribution.

Multivariate normal distribution

Definition

Z is a standard normal p-r.v. (short $Z \sim \mathcal{N}_p(0, I_p)$) if its components $Z_i \overset{i.i.d.}{\sim} N(0,1)$.

Recall that the joint density of independent variables, is the product of individual densities. Therefore it can be easily deduced that

$$x \mapsto f^Z(x) = \left(\frac{1}{2\pi}\right)^{\frac{p}{2}} \exp(-\|x\|^2/2).$$

Moreover, $EZ = 0$ and $\mathrm{Var}(Z) = I_p$.

Multivariate normal distribution

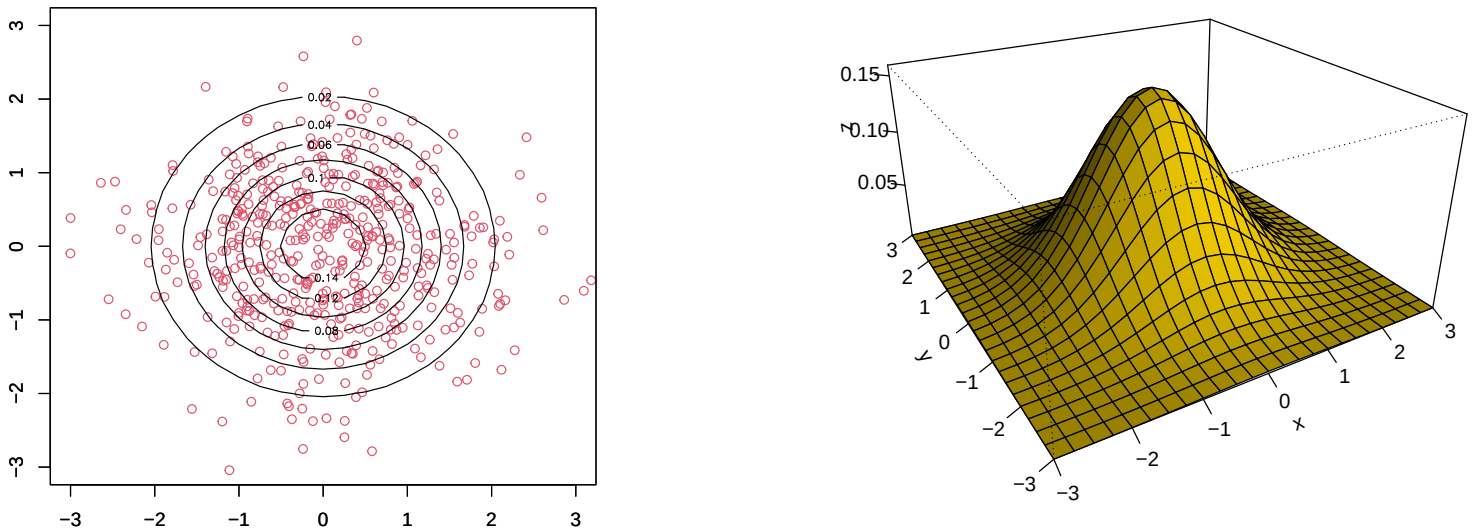


Figure 1: 500 i.i.d. observations from $\mathcal{N}_2(0, I_2)$ with density contours (left) and the density (right)

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Multivariate normal distribution

Definition

X has a p-variate normal distribution if for some vector $\mu \in \mathbb{R}^p$ and some matrix A $(p \times q)$ we have $X \overset{d}{\sim} AZ + b$, where $Z \sim \mathcal{N}_q(0, I_q)$.

We now list a few important properties:

Multivariate normal distribution

(i) $\mu := EX = b$ and $\Sigma := \mathrm{Var}(X) = AA'$.

It can be shown that the normal distribution depends only on μ and Σ.

We write: $X \sim \mathcal{N}_p(\mu, \Sigma)$.

Multivariate normal distribution

(ii) If $X \sim \mathcal{N}_p(\mu, \Sigma)$, then $Y = BX + c$ ($c \in \mathbb{R}^d$ and $B \in \mathbb{R}^{d \times p}$) is a d-variate normal $Y \sim \mathcal{N}_d(c + B\mu, B\Sigma B')$.

Exercise: it follows that every sub-vector of a multivariate normal is multivariate normal.

Multivariate normal distribution

(iii) If $\Sigma > 0$ is a symmetric $p \times p$-matrix $X \sim \mathcal{N}(\mu, \Sigma)$, then X has a density function:

$$x \mapsto f^X(x) = \left(\frac{1}{2\pi}\right)^{\frac{p}{2}} \frac{1}{|\Sigma|^{\frac{1}{2}}} \exp(-(x-\mu)'\Sigma^{-1}(x-\mu)/2).$$

Multivariate normal distribution

$$\mu = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix}.$$

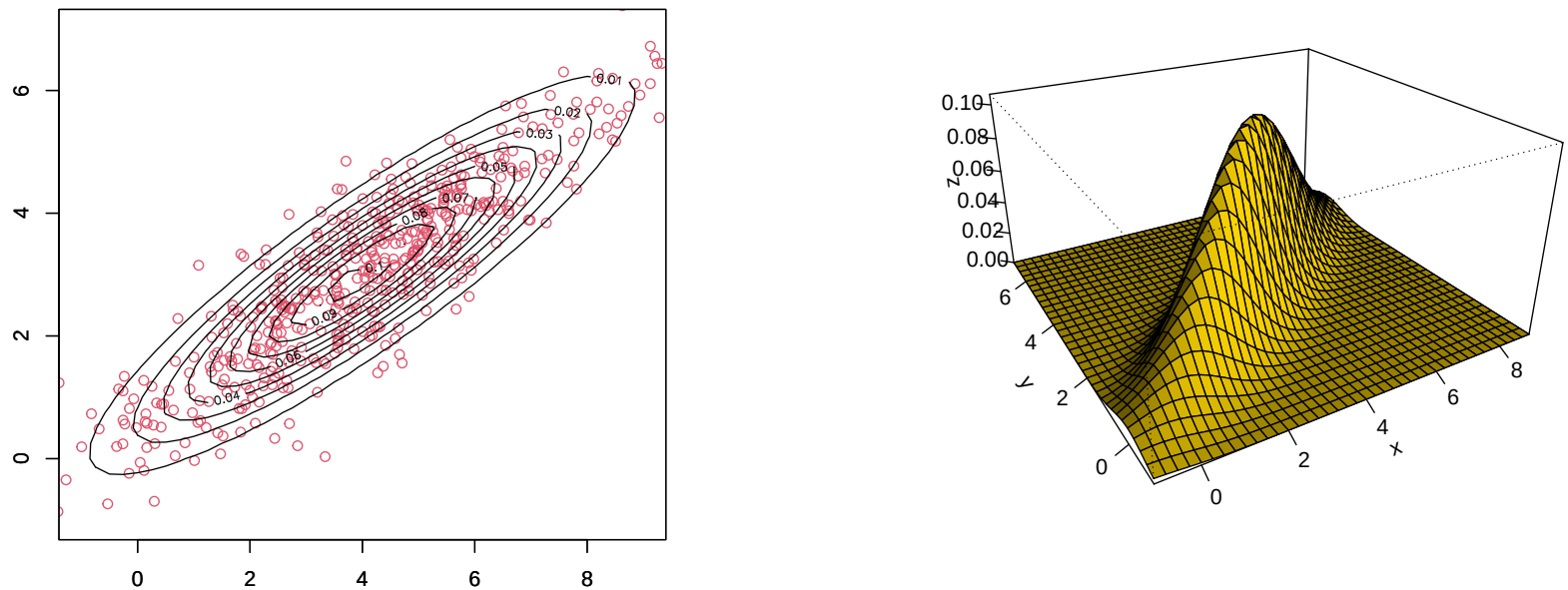


Figure 2: 500 i.i.d. observations from $\mathcal{N}_2(\mu, \Sigma)$ with density contours (left) and the density (right)

Multivariate normal distribution

For better understanding of the distribution, let's look closer at the contours:

$$\{x \colon f^X(x) = c\}$$

if and only if (iff)

$$\{x \colon (x-\mu)'\Sigma^{-1}(x-\mu) = c'\}.$$

These are ellipsoids with center μ. The shape of the ellipsoids is determined by Σ. How, we are going to see later.

Multivariate normal distribution

To go further, we need a little bit of linear algebra.

Remember that $\Sigma = \Sigma'$ and $v'\Sigma v \geq 0$. Linear algebra teaches us that in this case we may write

$$\Sigma = O\Lambda O' \quad \text{(this is the so-called spectral decomposition)}$$

where

- O is an orthogonal matrix ($OO' = O'O = \mathrm{Id}$).
  Hence: the inverse of an orthogonal matrix is its transpose.
- $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$ is a diagonal matrix with $\lambda_i \geq 0$.

Multivariate normal distribution

In our example:

$$\Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix} = \underbrace{\begin{pmatrix} 0.84 & -0.54 \\ 0.54 & 0.84 \end{pmatrix}}_{O} \underbrace{\begin{pmatrix} 6.92 & 0 \\ 0 & 0.32 \end{pmatrix}}_{\Lambda} \underbrace{\begin{pmatrix} 0.84 & 0.54 \\ -0.54 & 0.84 \end{pmatrix}}_{O'}$$

- The columns of O are orthogonal (scalar products are 0).
- The columns of O have length equal to 1.

Hence: columns of O are orthonormal vectors.

Multivariate normal distribution

Orthogonal matrices define rotations:

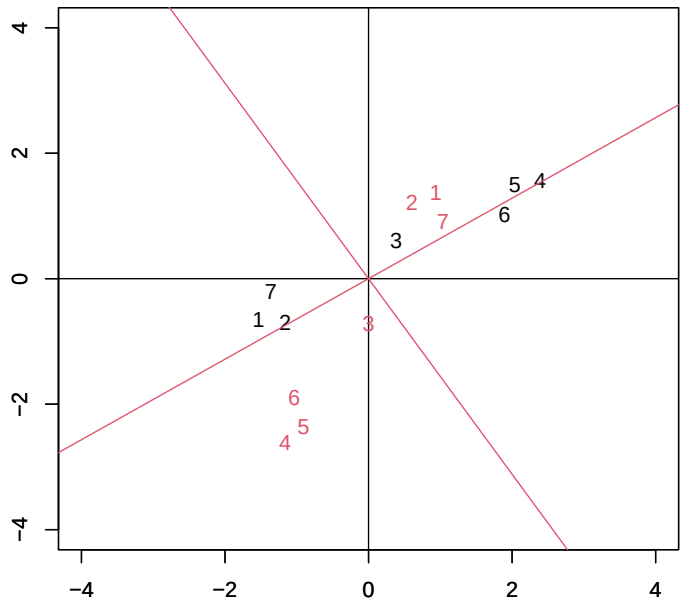


Figure 3: We see $x_1, \ldots, x_7$ and $Ox_1, \ldots, Ox_7$.

Multivariate normal distribution

How do we compute O and Λ?

The columns of O are the eigenvectors of Σ and the diagonal elements of Λ are the eigenvalues of Σ:

$$O = (e_1, \ldots, e_p) \quad \text{with} \quad \Sigma e_k = \lambda_k e_k.$$

As we have mentioned before: $e_k' e_l = \begin{cases} 0 & k \neq l; \\ 1 & k = l. \end{cases}$

Multivariate normal distribution

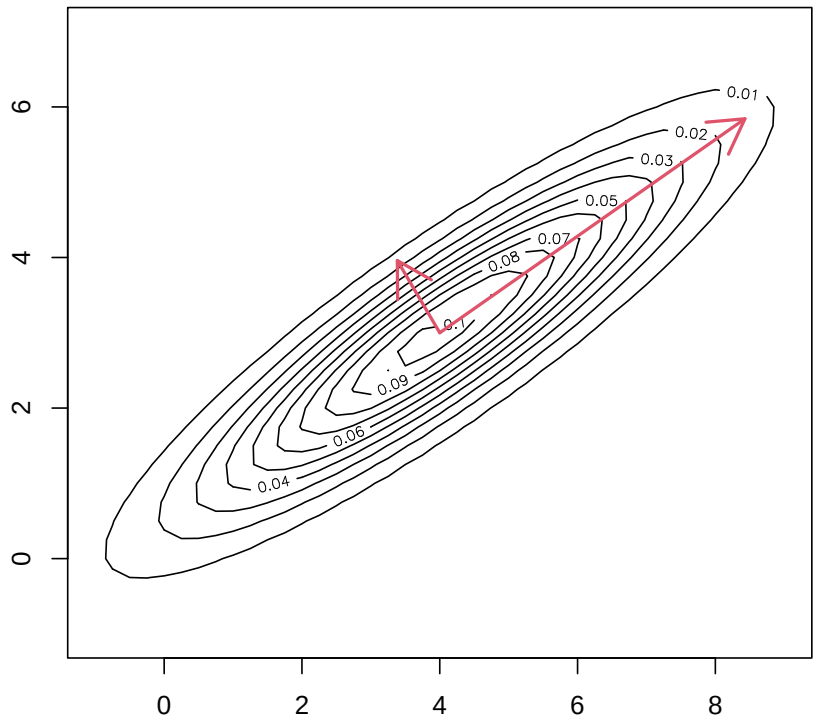


Figure 4: The eigenvectors are giving the main axes of the ellipsoides defining the contours. The length of the axes determined by e_k is proportional to $\sqrt{\lambda_k}$.

Multivariate normal distribution

If we want to measure how far a multivariate normal data point is from the center, it makes now sense to use the following distance:

Definition

The Mahalanobis distance between x and μ in the metric associated to Σ is defined as

$$d_\Sigma(x, \mu) = \sqrt{(x - \mu)' \Sigma^{-1} (x - \mu)}.$$

Multivariate normal distribution

We have learned how to determine the shape of the contours/data cloud using the spectral decomposition of Σ.

We can also use it to create a multivariate normal vector with a given Σ:

(iii) Let $\mu \in \mathbb{R}^p$ and $\Sigma \in \mathbb{R}^{p \times p}$ is symmetric and non-negative definite (we write $\Sigma \geq 0$), then there exists $X \sim \mathcal{N}_p(\mu, \Sigma)$.

Proof: Set $X = \mu + \Sigma^{1/2} Z$ with $Z \sim \mathcal{N}_p(0, I_p)$ and

$$\Sigma^x = O \Lambda^x O' \quad \text{for any } x \geq 0.$$

Note: if all $\lambda_k > 0$, we can extend the definition to all $x \in \mathbb{R}$.

Multivariate normal distribution

(vi) On the other hand, if $X \sim \mathcal{N}(\mu, \Sigma)$ and if $\Sigma > 0$, then

$$\Sigma^{-1/2}(X - \mu) \sim \mathcal{N}_p(0, I_p).$$

Hence, we can standardize X.

Multivariate normal distribution

Proposition

If $X \sim \mathcal{N}_p(\mu, \Sigma)$ *with* $\Sigma > 0$, $d^2_\Sigma(X, \mu) \sim \chi^2_p$.

Proof.

$$d^2_\Sigma(X, \mu) = (X - \mu)'\Sigma^{-1}(X - \mu) = \|\Sigma^{-1/2}(X - \mu)\|^2.$$

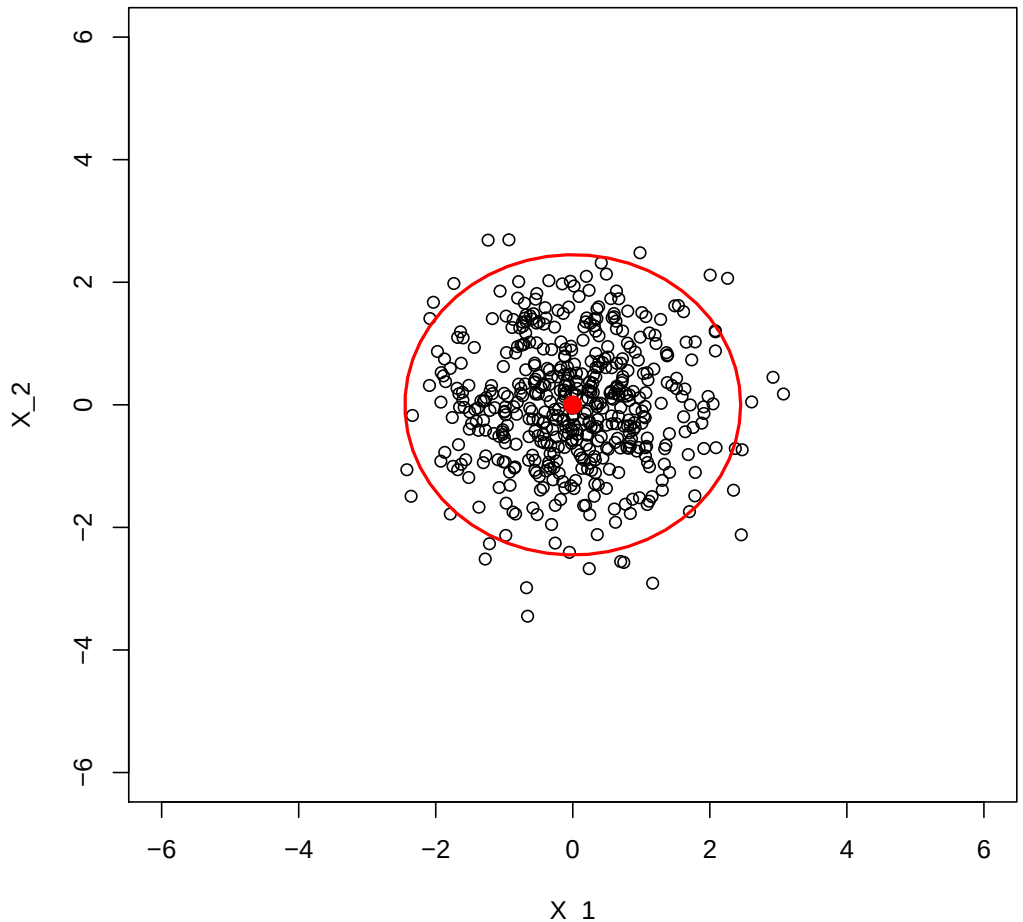
□

Multivariate normal distribution

Consequently, the probability mass contained in the ellipsoide $E_{1-\alpha} := \{y \in \mathbb{R}^p \mid d^2_\Sigma(y, \mu) \leq \chi^2_{p;1-\alpha}\}$ (where $\chi^2_{p;1-\alpha}$ is the $1-\alpha$ quantile of the χ^2_p distribution) is $1-\alpha$. We call it a $(1-\alpha) \times 100\%$ tolerance zone.

Multivariate normal distribution

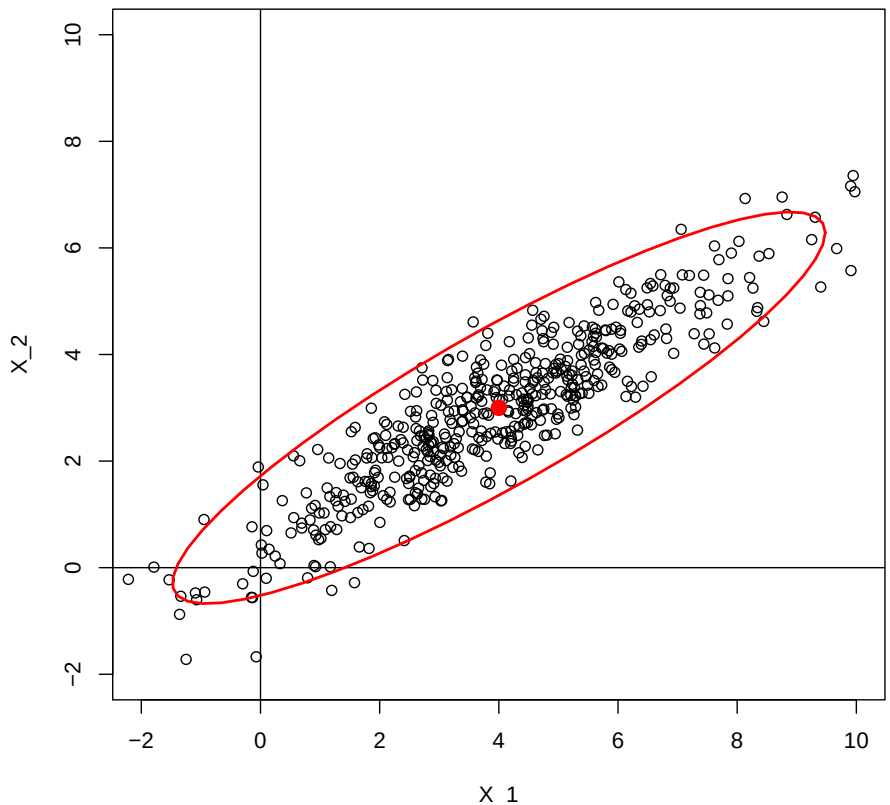
500 i.i.d. observations from $\mathcal{N}_2(0, I_2)$ and the tolerance zone $E_{.95}$.



Multivariate normal distribution

500 i.i.d. observations from $\mathcal{N}_2(\mu, \Sigma)$ and the tolerance zone $E_{.95}$ where

$$\mu = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad \text{et} \quad \Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix}.$$



Multivariate normal distribution

Let X_1 be p_1-r.v. and X_2 a p_2-r.v. We know that

- $X_1 \perp\!\!\!\perp X_2 \Rightarrow \mathrm{Cov}(X_1, X_2) = 0$, but
- the other way round is generally false.

Multivariate normal distribution

(v) Let $X = (X_1', X_2')' \sim \mathcal{N}_p(\mu, \Sigma)$, where X_i is a p_i-r.v. $(i = 1, 2)$ and where

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

Then $X_1 \perp\!\!\!\perp X_2$ if and only if $\Sigma_{12} = 0$.

Fisher's lemma

The following multivariate version of Fisher's lemma holds:

Proposition

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}_p(\mu, \Sigma)$. Then

(a) $\bar{X} \sim \mathcal{N}_p(\mu, \frac{1}{n}\Sigma)$,

(b) $(n-1)S \sim W_p(n-1, \Sigma)$, *and*

(c) $\bar{X} \perp\!\!\!\perp S$.

[$W_p(n-1, \Sigma)$ in (b) will be defined below.]

Fisher's lemma

Ad (b):

Definition

Let V be a $p \times p$ random matrix. Then

- *V has a Wishart distribution with m degrees of freedom (notation: $V \sim W_p(m)$) if $V \stackrel{\mathcal{D}}{=} \sum_{i=1}^{m} Z_i Z_i'$, where the Z_i are i.i.d. $\mathcal{N}_p(0, I_p)$.*
- *V has a Wishart distribution with parameter Σ (a $p \times p$ symmetric and > 0 matrix) with m degrees of freedom (notation: $V \sim W_p(m, \Sigma)$) if $V \stackrel{\mathcal{D}}{=} \sum_{i=1}^{m} X_i X_i'$, where the X_i are i.i.d. $\mathcal{N}_p(0, \Sigma)$.*

Fisher's lemma

Remarks

- $$V \sim W_1(m) \Leftrightarrow V \sim \chi^2_m$$

  and

  $$V \sim W_1(m, \sigma^2) \Leftrightarrow V \sim \sigma^2 \chi^2_m;$$

- Letting $X_i = \Sigma^{1/2} Z_i$ it becomes clear that

  $$V \sim W_p(m, \Sigma) \Leftrightarrow V \sim \Sigma^{1/2} V_0 \Sigma^{1/2},$$

  where $V_0 \sim W_p(m)$.